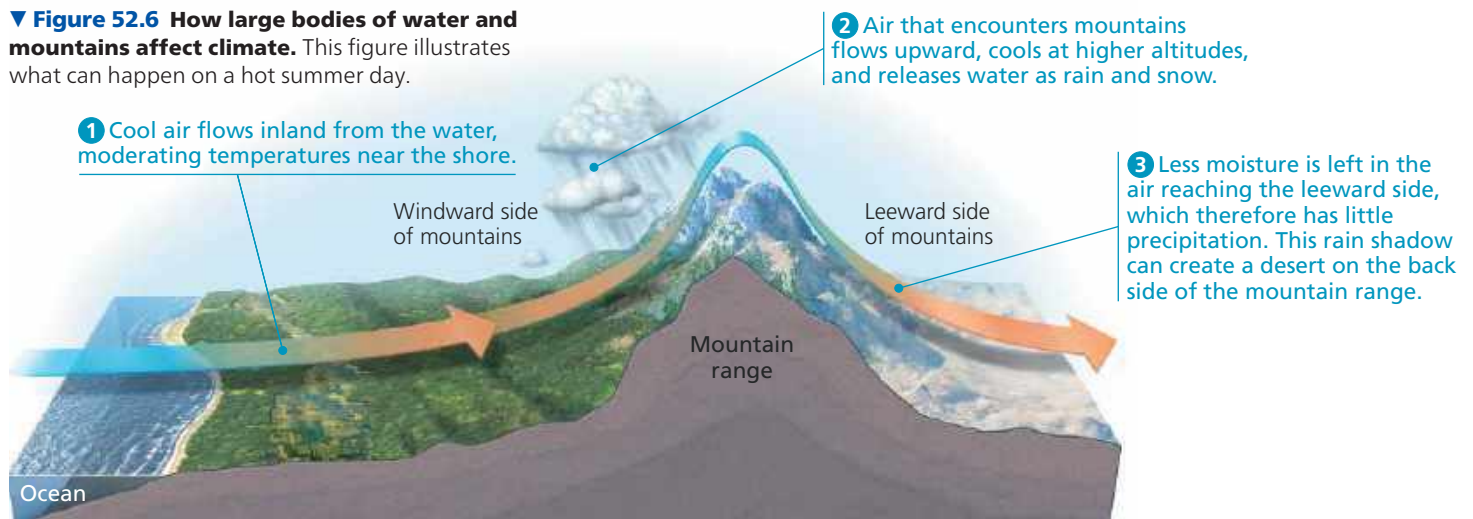


▼ **Figure 52.6** How large bodies of water and mountains affect climate. This figure illustrates what can happen on a hot summer day.



Effects of Vegetation on Climate

Climate affects where terrestrial organisms can live, but organisms can also affect climate. This is especially true of forests, which can alter the climate at local and even regional scales.

When viewed from above, a forest is darker in color than a desert or grassland. As a result, a forest absorbs more (and reflects less) solar energy than does a desert or grassland, thereby contributing to a warming of Earth's surface in forested areas. This warming effect, however, is more than offset by *transpiration*, the evaporative loss of water from a plant that cools the plant's surface—similar to how your body is cooled when you sweat. The evaporative loss of water is much greater in forests than in other ecosystems, causing forests to affect climate in two key ways: Forests reduce Earth's surface temperature and they increase precipitation rates (**Figure 52.7**). As documented in regions around the world, the climate becomes hotter and drier in areas where humans have cut down large forests and becomes cooler and wetter where humans have restored large forests.

Microclimate

At an even smaller scale is the **microclimate**, very fine, localized patterns in climatic conditions. Many features in the environment influence microclimate by casting shade, altering evaporation from soil, or changing wind patterns. For example, forest trees often moderate the microclimate below them. Cleared areas therefore typically experience greater temperature extremes than the forest interior because of greater solar radiation and wind currents that arise from the rapid heating and cooling of open land. Within a forest, low-lying ground is usually wetter than higher ground and tends to be occupied by different tree species. A log or large stone can shelter organisms such as salamanders, worms, and insects, buffering them from the extremes of temperature and moisture.

Every environment on Earth exhibits small-scale differences in chemical and physical attributes, such as temperature, light, water, and nutrients. Later, we'll examine how these **abiotic**, or nonliving, factors influence the distribution and abundance

▼ **Figure 52.7** How cutting down a forest affects regional climate.

